Practical Exercise 1 (not assessed)

1. Aims and objectives

Familiarise yourselves with the environment we will use in our assignments, that is, Microsoft Visual Studio and Microsoft Visual C++, and try to compile and run the program supplied with this guideline.

2. Tasks

2.1. Start Microsoft Visual Studio

Start Microsoft Visual Studio from the Start Menu. Create a new project by selecting:

`[File] -> [New] -> [Project…]`,

then choose in **`New Project`** dialog:

`[Visual C++] -> [Win32] -> [Win32 Console Application]`.

Now in the text field labeled **`Name:`** type in the name of your project, say, `Practical1`. In the following window click the button **`Finish`**.

2.2. Configure your project

Copy the program supplied with this guideline into the created source file `Practical1.cpp`. Before you can run the program you must configure your project as follows:

`[Project] -> [Project properties] -> [Configuration properties] -> [C\C++] -> [Code generation] -> [Runtime Library] -> Multi-threaded/MT`,

and,

`[Project] -> [Project properties] -> [Configuration properties] -> [C\C++] -> [Precompiled Headers] -> [Precompiled Header] -> Not Using Precompiled Headers`.

2.3. Debug and execute

Once you have done the above settings, try the following:

- Press **F5** to build executable code;
- Press **F10** to start single stepping in a debugger.

Yellow arrow will appear in the main window pointing out the instruction to be executed. Several new windows will appear. Familiarise yourselves with the windows. You can switch them on and off. Try different executions modes - **step into, step over, step out of** - by pressing suitable buttons on the top tool bar. In the window **`Autos`** you can see how the execution of instruction changes the content of registers and variables.

3. The program

Copy-and-paste the code below into the CPP file generated by your preferred version of Visual Studio.

```
char message1[] = "Give me a first number: ";
char message2[] = "\nGive me a second number: ";
char message3[] = "\nThe numbers are equal! \n";
char message4[] = "\nThe numbers are not equal! \n";
char message5[] = "Type in any integer and press RETURN key to finish: ";
char format[] = "%d"; // format string for the scanf function

int first;
int second;
int end;
_asm {
    lea eax, message1     ;
    push eax          ; PRINTING the FIRST MESSAGE
    call printf           ;
    add esp,4             ;

    lea eax,first         ;
    push eax          ;
    lea eax,format        ; READING the FIRST NUMBER
    push eax          ;
    call scanf_s      ;
    add esp,8             ;

    lea eax, message2     ;
    push eax          ; PRINTING the SECOND MESSAGE
    call printf           ;
    add esp,4             ;

    lea eax,second        ;
    push eax          ;
    lea eax,format        ; READING the SECOND NUMBER
    push eax          ;
    call scanf_s      ;
    add esp,8             ;

    mov eax,first         ;
    sub eax,second        ; COMPARE TWO NUMBERS, JUMP to suitable section
    jnz nequal            ;
```

```
equal: lea eax, message3 ;
    push eax         ;
    call printf          ; PRINTING "Numbers are equal"
    add esp,4            ;
    jmp finish           ; JUMP to finish

nequal: lea eax, message4;
    push eax         ; PRINTING "Numbers are not equal"
    call printf          ;
    add esp,4            ;

finish: lea eax, message5;
    push eax         ; PRINTING the FIFTH MESSAGE
    call printf          ;
    add esp,4            ;

    lea eax, end     ;
    push eax         ;
    lea eax, format      ;
    push eax         ;
    call scanf_s     ;
    add esp, 8           ;
}
return 0;
```